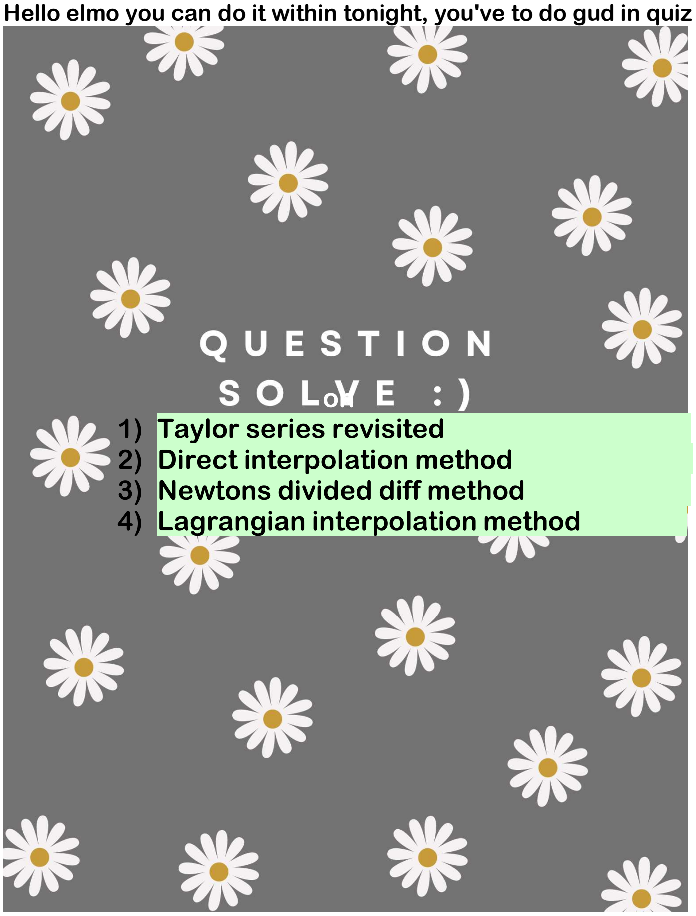
Question solve for quiz-2

Wednesday, February 21, 2024 9:18 PM

Hello elmo you can do it within tonight, you've to do gud in quiz no matter what's goin on in life or if anyone is there for you in life or not.



1) Taylor series revisited
2) Direct interpolation method
3) Newtons divided diff method
4) Lagrangian interpolation method

- [x] 1. SWE 20 batch quiz question
- [x] 2. SWE 20 Mid question
- [ ] 3. SWE 20 Final question
- [x] 4. CSE 19 Quiz question
- [x] 5. CSE 19 Mid question
- [x] 6. CSE 19 Final question
- [x] 7. SWE 19 Question(whatever I can find)
- [x] 8. SWE 19 MID
- [x] 9. SWE 19 FINAL
- [x] 10. SWE 19 QUIZ 2
- [x] 11. CSE 18 Quiz question
- [x] 12. CSE 18 MID question
- [x] 13. CSE 18 Final question
- [x] 14. EXTRA QUESTION OF YEAR 17-18
- [ ] 15. Book question

SOLUTION

FIRST WORK IN THE MORNING
GIVE AN EXAM ON THE QUIZ QUESTION OF SWE 20 WATCHING THE TIME (MUST)

- [x] 1. SWE 20 batch quiz question

1. The Taylor series for a function $f(x)$ is — 6 (CO1) (PO2)

$$f(x+h) = f(x) + f'(x)\frac{h}{1!} + f''(x)\frac{h^2}{2!} + f'''(x)\frac{h^3}{3!} + f''''(x)\frac{h^4}{4!} + f'''''(x)\frac{h^5}{5!} + \ldots$$

Using the Taylor series (with at least the first 4 terms), derive the Maclaurin series of —

(i) $\ln(1+x)$

(ii) $\ln(1-x)$

Then use these two series to determine the Maclaurin series of $\ln\left(\frac{1+x}{1-x}\right)$.

Solution:

The Maclaurin series of a transcendental function is simply a Taylor series around the point $x = 0$.

We know, $\frac{d}{dx}(\ln(x)) = \frac{1}{x}$, and $\frac{d}{dx}(x^n) = nx^{n-1}$.

(i) Now, for $f(x) = \ln(1+x)$, let's calculate the derivatives at the point $x = 0$.

$f(x) = \ln(1+x)$, $f(0) = \ln(1+0) = \ln(1) = 0$; $f'(x) = \frac{1}{1+x}$, $f'(0) = \frac{1}{1+0} = 1$;

$f''(x) = -\frac{1}{(1+x)^2}$, $f''(0) = -\frac{1}{(1+0)^2} = -1$; $f'''(x) = \frac{2}{(1+x)^3}$, $f'''(0) = \frac{2}{(1+0)^3} = 2$;

Using the first 4 terms of the Taylor series,

$$f(x+h) = f(x) + f'(x)\frac{h}{1!} + f''(x)\frac{h^2}{2!} + f'''(x)\frac{h^3}{3!} + \ldots$$

For $x = 0$,

$$f(0+h) = f(0) + f'(0)\frac{h}{1!} + f''(0)\frac{h^2}{2!} + f'''(0)\frac{h^3}{3!} + \ldots$$

$$\Rightarrow f(h) = 0 + h - \frac{h^2}{2!} + \frac{2h^3}{3!} + \ldots$$

$$\Rightarrow f(x) = 0 + x - \frac{x^2}{2!} + \frac{2x^3}{3!} + \ldots$$

$$\Rightarrow \ln(1+x) = 0 + x - \frac{x^2}{2!} + \frac{2x^3}{3!} + \ldots$$

$$\Rightarrow \ln(1+x) = x - \frac{x^2}{2!} + \frac{2x^3}{3!} + \ldots$$

(ii) Short version: Here,

$$\begin{aligned} f(x) &= \ln(1-x) \\ &= \ln(1+(-x)) \\ &= (-x) - \frac{(-x)^2}{2!} + \frac{2(-x)^3}{3!} + \dots \qquad \text{[using the Maclaurin series of } \ln(1+x) \text{ from (i)]} \\ &= -x - \frac{x^2}{2!} - \frac{2x^3}{3!} + \dots \end{aligned}$$

Long version: Now, for $f(x) = \ln(1-x)$, let's calculate the derivatives at the point $x = 0$.

$f(x) = \ln(1-x)$, $f(0) = \ln(1-0) = \ln(1) = 0$; $f'(x) = -\dfrac{1}{1-x}$, $f'(0) = -\dfrac{1}{1-0} = -1$;

$f''(x) = -\dfrac{1}{(1-x)^2}$, $f''(0) = -\dfrac{1}{(1-0)^2} = -1$;

$f'''(x) = -\dfrac{2}{(1-x)^3}$, $f'''(0) = -\dfrac{2}{(1-0)^3} = -2$;

Using the first 4 terms of the Taylor series,

$$f(x+h) = f(x) + f'(x)\frac{h}{1!} + f''(x)\frac{h^2}{2!} + f'''(x)\frac{h^3}{3!} + \dots$$

For $x = 0$,

$$\begin{aligned} f(0+h) &= f(0) + f'(0)\frac{h}{1!} + f''(0)\frac{h^2}{2!} + f'''(0)\frac{h^3}{3!} + \dots \\ \Rightarrow f(h) &= 0 - h - \frac{h^2}{2!} - \frac{2h^3}{3!} + \dots \\ \Rightarrow f(x) &= 0 - x - \frac{x^2}{2!} - \frac{2x^3}{3!} + \dots \\ \Rightarrow \ln(1-x) &= 0 - x - \frac{x^2}{2!} - \frac{2x^3}{3!} + \dots \\ \Rightarrow \ln(1-x) &= -x - \frac{x^2}{2!} - \frac{2x^3}{3!} + \dots \end{aligned}$$

We know, $\ln\left(\frac{a}{b}\right) = \ln(a) - \ln(b)$.

So,

$$\begin{aligned} \ln\left(\frac{1+x}{1-x}\right) &= \ln(1+x) - \ln(1-x) \\ &= \left(x - \frac{x^2}{2!} + \frac{2x^3}{3!} + \dots\right) - \left(-x - \frac{x^2}{2!} - \frac{2x^3}{3!} + \dots\right) \\ &= 2x + \frac{4x^3}{3!} + \dots \end{aligned}$$

Rubric:

- 2.5 points for the correct series of $\ln(1+x)$.
- 2.5 points for the correct series of $\ln(1-x)$.
- 1 point for the correct series of $\ln\left(\dfrac{1+x}{1-x}\right)$.

2. Compare and contrast the following interpolation methods– ***Direct*** method, ***Newton's Divided Difference*** method, and ***Lagrangian*** method. 2 (CO3) (PO2, PO4)

Solution:

Any 2 valid comparisons will suffice. Some comparisons can be–

Criteria	Direct	Newton's Divided Difference	Lagrangian
Interpolant Polynomial Representation	$f_n(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$	$f_n(x) = b_0 + b_1(x-x_0) + \dots + b_n(x-x_0)(x-x_1)\dots(x-x_{n-1})$	$f_n(x) = \sum_{i=0}^{n}\left(\prod_{j=0, j\neq i}^{n} \frac{x-x_j}{x_i-x_j}\right) f(x_i)$
Computational Complexity	Depends on the algorithm used to solve the equations. It may be $O(n^3)$, $O(n^3\log(\lVert A\rVert + \lVert b\rVert))$, or $O((n+1)!)$.	Using memoized dynamic programming, we get a complexity of $O(n^2)$.	The naïve implementation is $O(n^2)$, but it can optimized further down to $O(n\log^2(n))$.
Property of the terms	Each term has a coefficient a_i and these coefficients are obtained by solving all the equations simultaneously.	Each term has a coefficient b_i and each of these coefficients can be obtained by solving one equation.	The polynomial is a weighted sum of the given functional values $f(x_i)$, and each term is a product of the Lagrangian weight function $L_i(x)$ and the functional value $f(x_i)$.
Vulnerable to the Runge Phenomenon?	Yes	Yes	Yes

Rubric:

- 1 point for each valid comparison.

3. The specific heat capacity C of a substance is defined as the amount of heat that is required to raise the temperature of unit mass of that particular substance by 1 degree. Suppose, you are conducting an experiment to determine how much heat is required to bring some volume of water to its boiling point. The values of specific heat capacity C of water that you have calculated at different temperature values T are shown in the table below – 4 (CO2) (PO3)

Temperature, T ($°C$)	Specific Heat, C ($Jkg^{-1}°C^{-1}$)
22	4181
42	4179
52	4186
82	4199
100	4217

Table 1: Specific heat C of water as a function of temperature T.

Determine the value of the specific heat at $T = 61°C$ using the ***Direct method*** of interpolation and a second order polynomial.

Solution:

For second order polynomial interpolation (also called quadratic interpolation), the equation for the specific heat would be –

$$C(T) = a_0 + a_1T + a_2T^2$$

Here, the order $n = 2$. Now, let's choose the nearest $n + 1 = 2 + 1 = 3$ points that bracket the point $T = 61°C$ and evaluate the polynomial. The 3 points are $T_0 = 42$, $T_1 = 52$, and $T_2 = 82$. The respective specific heat values are $C(T_0) = 4179$, $C(T_1) = 4186$, and $C(T_2) = 4199$. Using these values, we obtain the following equations –

$$C(42) = a_0 + a_1 \times 42 + a_2 \times (42)^2 = 4179$$
$$C(52) = a_0 + a_1 \times 52 + a_2 \times (52)^2 = 4186$$
$$C(82) = a_0 + a_1 \times 82 + a_2 \times (82)^2 = 4199$$

Writing these equations in matrix form, we have –

$$\begin{bmatrix} 1 & 42 & 1764 \\ 1 & 52 & 2704 \\ 1 & 82 & 6724 \end{bmatrix} \cdot \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 4179 \\ 4186 \\ 4199 \end{bmatrix} \Rightarrow \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 1 & 42 & 1764 \\ 1 & 52 & 2704 \\ 1 & 82 & 6724 \end{bmatrix}^{-1} \cdot \begin{bmatrix} 4179 \\ 4186 \\ 4199 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} \frac{533}{50} & -\frac{287}{25} & \frac{91}{50} \\ -\frac{67}{200} & \frac{31}{75} & -\frac{47}{600} \\ \frac{1}{400} & -\frac{1}{300} & \frac{1}{1200} \end{bmatrix} \cdot \begin{bmatrix} 4179 \\ 4186 \\ 4199 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 4135 \\ 1.3267 \\ -6.6667 \times 10^{-3} \end{bmatrix}$$

Hence,

$$C(T) = 4135 + 1.3267T - 6.6667 \times 10^{-3}T^2; \qquad 42 \le T \le 82$$

At the point $T = 61$, we get,

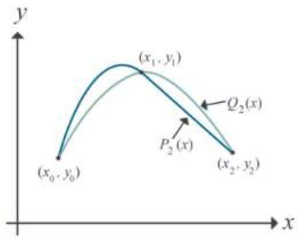
$$C(61) = 4135 + 1.3267 \times 61 - 6.6667 \times 10^{-3} \times (61)^2 = 4191.2Jkg^{-1}°C^{-1}$$

Rubric:

- 1 point for the correct polynomial representation.
- 1 point for choosing the correct nearest points.
- 1 point for the correct coefficient values.
- 1 point for the correct answer.

4. Prove that a polynomial of degree n or less that passes through $n+1$ data points is unique. 3 (CO1) (PO2)

Solution:



Proof: Let us use proof by contradiction. If the polynomial is not unique, then at least two polynomials of order n or less pass through the $n+1$ data points.

Assume two polynomials $P_n(x)$ and $Q_n(x)$ go through $n+1$ data points (x_0, y_0), (x_1, y_1), ..., (x_n, y_n). Then,

$$R_n(x) = P_n(x) - Q_n(x)$$

Since $P_n(x)$ and $Q_n(x)$ pass through all the $n+1$ data points, we can say,

$$P_n(x_i) = Q_n(x_i) \quad \forall i \in \{0, \ldots, n\}$$

Hence

$$R_n(x_i) = P_n(x_i) - Q_n(x_i) \quad \forall i \in \{0, \ldots, n\}$$
$$\Rightarrow R_n(x_i) = 0$$

The nth order polynomial $R_n(x)$ has $n+1$ zeros. A polynomial of order n can have $n+1$ zeros only if it is identical to a zero polynomial, that is,

$$R_n(x) \equiv 0$$
$$\Rightarrow P_n(x) \equiv Q_n(x)$$

So, $P_n(x)$ and $Q_n(x)$ must be the same unique polynomial, which contradicts our initial assumption. □ Q.E.D.

Rubric:

- 1 point for correctly defining $P_n(x)$, $Q_n(x)$, and $R_n(x)$.
- 1 point for showing $R_n(x)$ is a zero polynomial.
- 1 point for the final conclusion.

- [x] **2. SWE 20 Mid question**

(imp question for taylor series)

b) The Taylor series for a function $f(x)$ is — 15 (CO2) (PO1)

$$f(x+h) = f(x) + f'(x)\frac{h}{1!} + f''(x)\frac{h^2}{2!} + f'''(x)\frac{h^3}{3!} + f''''(x)\frac{h^4}{4!} + f'''''(x)\frac{h^5}{5!} + \ldots$$

Using the Taylor series (with at least the first 5 terms), derive the Maclaurin series of

$$f(x) = \sec(x)$$

Solution:

The Maclaurin series of a transcendental function is simply a Taylor series around the point $x = 0$.

We know, $\frac{d}{dx}(\sec(x)) = \sec(x)\tan(x)$, and $\frac{d}{dx}(u(x)v(x)) = u(x)\frac{d}{dx}(v(x)) + v(x)\frac{d}{dx}(u(x))$.
Now, for $f(x) = \sec(x)$, let's calculate the derivatives at the point $x = 0$.

$$f(x) = \sec(x), \quad f(0) = 1$$
$$f'(x) = \sec(x)\tan(x), \quad f'(0) = 0$$
$$f''(x) = \sec^3(x) + \sec(x)\tan^2(x), \quad f''(0) = 1$$
$$f'''(x) = 5\sec^3(x)\tan(x) + \sec(x)\tan^3(x), \quad f'''(0) = 0$$
$$f''''(x) = 5\sec^5(x) + 18\sec^3(x)\tan^2(x) + \sec(x)\tan^4(x), \quad f''''(0) = 5$$

Using the first 5 terms of the Taylor series,

$$f(x+h) = f(x) + f'(x)\frac{h}{1!} + f''(x)\frac{h^2}{2!} + f'''(x)\frac{h^3}{3!} + f''''(x)\frac{h^4}{4!} + \ldots$$

For $x = 0$,

$$f(0+h) = f(0) + f'(0)\frac{h}{1!} + f''(0)\frac{h^2}{2!} + f'''(0)\frac{h^3}{3!} + f''''(0)\frac{h^4}{4!} + \ldots$$
$$\Rightarrow f(h) = 1 + 0 + \frac{h^2}{2!} + 0 + \frac{5h^4}{4!} + \ldots$$
$$\Rightarrow f(x) = 1 + \frac{x^2}{2!} + \frac{5x^4}{4!} + \ldots$$
$$\Rightarrow \sec(x) = 1 + \frac{x^2}{2!} + \frac{5x^4}{4!} + \ldots$$
$$\Rightarrow \sec(x) = 1 + \frac{x^2}{2} + \frac{5x^4}{24} + \ldots$$

Rubric:

- 1 point for defining $f(x)$ and finding each i-th order derivative $f^i(x)$.
- 1 point for correct values of $f(0)$ and each $f^i(0)$.
- 5 points for deriving the correct series.

c) The Maclaurin series of the exponential function e^x is —

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \ldots$$

i. Why is the exponential function e^x categorized as a *transcendental function*? 1 (CO4) (PO1)

Solution:

The exponential function e^x cannot be expressed using a finite sequence of terms or algebraic operations which is why it falls under the umbrella of transcendental functions.

Rubric:

- 1 point for the correct reason.

ii. Using the *Remainder Theorem*, establish the bounds of the truncation error in the representation of $e^{1.69}$ if only the first 5 terms of the series are used. 4 (CO2) (PO1)

Solution:
If we are using 5 terms of the Taylor series, the remainder is given by ($x = 0$, $h = 1.69$, $n = 4$)

$$R_4(0 + 1.69) = \frac{1.69^{(4+1)}}{(4+1)!} f^{(4+1)}(c) = \frac{1.69^5}{5!} f^5(c) = \frac{1.69^5 e^c}{120}$$

Since,

$$x < c < x + h$$
$$0 < c < 0 + 1.69$$
$$0 < c < 1.69$$

The error bound is between,

$$\frac{1.69^5 e^0}{120} < R_4(1.69) < \frac{1.69^5 e^{1.69}}{120}$$
$$0.1148 < R_4(1.69) < 0.6226$$

Rubric:

- 1 point for finding $R_4(1.69)$ in terms of c.
- 1.5 points for correct lower-bound.
- 1.5 points for correct upper-bound.

b) But is there any way to know the bounds of this error other than calculating it directly? Yes.

We know that

$$f(x+h) = f(x) + f'(x)h + \ldots + f^{(n)}(x)\frac{h^n}{n!} + R_n(x+h)$$

where

$$R_n(x+h) = \frac{(h)^{n+1}}{(n+1)!} f^{(n+1)}(c),\ x < c < x+h, \text{and}$$

c is some point in the domain $(x, x+h)$. So, in this case, if we are using four terms of the Taylor series, the remainder is given by $(x = 0, n = 3)$

3. a) Differentiate between *Interpolation* and *Extrapolation*. With suitable real-world examples, briefly write when each of the methods is applicable. 4 (CO3) (PO1)

Solution:

Any 2 valid conceptual differences will suffice. There should be at least 1 valid real-world example that is used to explain the applicability. Some differences are —

Criteria	Interpolation	Extrapolation
Position of newly estimated points	Within the distribution of the known data points.	Outside the distribution of the known data points.
Likelihood of proper estimation	Has a greater likelihood of obtaining a valid estimate, since the pattern of the data points on either side of the newly estimated point is known.	Has a lesser likelihood of obtaining a valid estimate, since it assumes that the observed trend continues for values of x outside the range of the given data points.
Applicability	• ***Computer Graphics*** – For constructing B-splines, smooth surfaces, etc. • ***Carpentry*** – For drawing curves on wooden surfaces using different apparatus *e.g.* the Spline tool. • ***Digital Image Processing*** – For increasing the resolution of images. • ***Electronics*** – For plotting transfer functions or calibration curves of electronic components *e.g.* Thermistors.	• ***Weather Forecasting*** – For predicting the temperature, humidity levels, and weather conditions of the future. • ***Stock Market*** – For estimating the future price of company shares and stocks. • ***Cricket*** – For hypothesizing the total runs that will be scored in the innings by the batting team.

b) Suppose, there is an unknown function $f(x)$ and you are given 3 points $(x_0, f(x_0))$, $(x_1, f(x_1))$, and $(x_2, f(x_2))$ through which $f(x)$ passes. Now, in order to approximate $f(x)$, you decide to use the *Newton's Divided Difference* method of interpolation and obtain a *quadratic interpolant* $f_2(x)$. Derive this 2nd order Newton's Divided Difference polynomial $f_2(x)$. 8 (CO4) (PO1)

Solution:

😑

Just click here.

Rubric:

- 1 point for writing the quadratic NDD interpolant $f_2(x)$ in the proper form.
- 1 point for correct derivation of b_0.
- 1 point for correct derivation of b_1.
- 4 points for correct derivation of b_2.
- 1 point for writing the correct expanded form of the quadratic NDD interpolant $f_2(x)$.

d) A thermistor, also known as thermal resistor, is a semiconductor type of resistor whose resistance is strongly dependent on temperature. To measure temperature using a thermistor, the manufacturers provide users with a Temperature (T) vs. Resistance (R) calibration curve. If we measure the resistance of the thermistor using an Ohmmeter, then we can refer to the aforementioned curve and determine the corresponding temperature value. Figure-1 and Table-2 portray multiple recorded observations involving a thermistor. 6 (CO2) (PO1)

Table 2: Temperature T of a thermistor as a function of its resistance R for Question-3(d).

Resistance, R (Ω)	Temperature, T ($^\circ C$)
1101	25.113
911.3	30.131
636	40.12
451.1	50.128

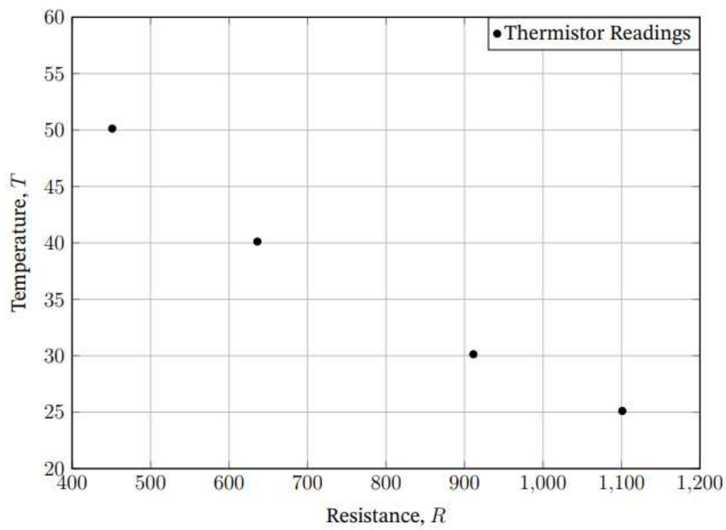


Figure 1: Temperature T vs. Resistance R for Question-3(d).

Determine the value of the temperature T when the thermistor resistance is $R = 754.8\Omega$ using the *Lagrangian method* of interpolation and a second order polynomial.

Solution:
The 2nd order Lagrangian interpolant is —

$$\begin{aligned} T(R) &= \sum_{i=0}^{2} L_i(R)T(R_i) \\ &= L_0(R)T(R_0) + L_1(R)T(R_1) + L_2(R)T(R_2) \end{aligned}$$

Given, $R = 754.8\Omega$, and the 3 nearest data points that bracket it are, $R_0 = 911.3$, $R_1 = 636$, and $R_2 = 451.1$. The corresponding T values are $T(R_0) = 30.131$, $T(R_1) = 40.12$, and $T(R_2) = 50.128$ respectively.
Now, let's calculate the Lagrangian weights.

$$L_0(R) = \prod_{\substack{j=0 \\ j\neq 0}}^{2} \frac{R - R_j}{R_0 - R_j} = \left(\frac{754.8 - 636}{911.3 - 636}\right) \times \left(\frac{754.8 - 451.1}{911.3 - 451.1}\right) = 0.284$$

$$L_1(R) = \prod_{\substack{j=0 \\ j\neq 1}}^{2} \frac{R - R_j}{R_0 - R_j} = \left(\frac{754.8 - 911.3}{636 - 911.3}\right) \times \left(\frac{754.8 - 451.1}{636 - 451.1}\right) = 0.933$$

$$L_2(R) = \prod_{\substack{j=0 \\ j\neq 2}}^{2} \frac{R - R_j}{R_0 - R_j} = \left(\frac{754.8 - 911.3}{451.1 - 911.3}\right) \times \left(\frac{754.8 - 636}{451.1 - 636}\right) = -0.218$$

Now that we have the weights, we can obtain the value of $T(754.8)$.

$$\begin{aligned} T(754.8) &= 0.284 \times T(911.3) + 0.933 \times T(636) + (-0.218) \times T(451.1) \\ &= 0.284 \times 30.131 + 0.933 \times 40.12 + (-0.218) \times 50.128 \\ &\approx 35.088°C \end{aligned}$$

3. **SWE 20 Final question**
4. **CSE 19 Quiz question**

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

Quiz no. 2 | **Summer Semester 2021-2022**

DURATION: 30 MIN | **FULL MARKS: 15**

Math 4641: Numerical Methods

Figures in the right margin indicate marks.

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1. Suppose you are given the following data points: 7 (CO2) (PO1)

x	0	0.5	1	1.5
$f(x)$	0	0.4794	0.8415	0.9975

Use the highest order polynomial possible for the given data points and find $f(0.6)$ using the Direct Interpolation Method.

2. Find the Maclaurin Series (showing at least first 4 terms) for $\ln(1+x)$ and $\ln(1-x)$ and hence find the Maclaurin Series for $\ln\left(\frac{1+x}{1-x}\right)$. **8** **(CO1)** **(PO2)**

5. CSE 19 Mid question

1. a) The upward velocity of a rocket is given as a function of time according to Table 1: 7 + 5 + 6 (CO2) (PO1)

Table 1: Time vs Velocity data

t (second)	Velocity (meter/second)
0	0
10	227.04
15	362.78
20	517.35
22.5	602.97
30	901.67

i. Determine the velocity of the rocket at $t = 11$ seconds using Second Order (quadratic) Lagrangian polynomial interpolation.

ii. Find the acceleration of the rocket at $t = 11$ seconds.

iii. Find the distance travelled by the rocket from $t = 10$ to $t = 14$ seconds.

b) Provide an analytical comparison among the solution approaches of Direct Interpolation, Newton's Divided Difference Interpolation and Lagrangian Interpolation. 7 (CO3) (PO2)

Show the absolute relative approximate error at every iteration.

b) Compute $f(4)$, $f'(4)$ and $f''(4)$ using Taylor Series given that $f(5) = -30$, $f'(5) = 10$, $f''(5) = 16$ and $f'''(5) = 6$ and all higher derivatives of $f(x)$ at $x = 5$ are zero. 12 (CO2) (PO1)

6. CSE 19 Final question

b) Suppose you are given the time vs velocity data for two separate missiles in Table 2. 18 (CO2) (PO1)

Table 2: Time vs Velocity data for two missiles for Question 3b)

t	V_1	V_2
0	0	0
10	227.04	152.35
15	362.78	316.62
20	517.35	502.71
22.5	602.97	620.25
30	901.67	978.87

Using Direct method of Interpolation with a third order polynomial, find the velocity of both missiles at $t = 21$ seconds. For solving the simultaneous equations, you are not allowed to use inverse matrix multiplication or Gaussian Elimination more than once.

7. SWE 19 Question(whatever I can find)

8. Swe 19 MID QUESTION

2. a) i. The Maclaurin series is a special case of Taylor series. Find the Maclaurin series for $\cos(x)$ starting from the general definition of Taylor Series. 4 (CO1) (PO2)

ii. Estimate the bounds of the truncation error in the representation of cos 2 using the Remainder Theorem, given that only the first 3 terms of the Maclaurin series are used. 6 (CO3) (PO2) used.

b) The upward velocity of a rocket is given as a function of time in Table 1:

Table 1: Velocity as function of time for Question 2.(b)

T (s)	V (ms^{-1})
0	0
10	227.04
15	362.78
20	517.35
22.5	602.97
30	901.67

Solve the following problems using the spline interpolation method with linear splines:

i. What is the velocity of the rocket at $t = 14$? 6 (CO2) (PO1)

ii. What is the acceleration of the rocket at $t = 21$? 4 (CO2) (PO1)

iii. What is the distance travelled by the rocket from $t = 14$ to $t = 21$? 5 (CO2) (PO1)

3. a) Find the Quadratic form of Newton's Divided Difference polynomial and using its pattern, find the Cubic form of Newton's Divided Difference polynomial. 9+3 (CO1) (PO2)

b) Compute $f(6)$, $f'(6)$ and $f''(6)$ given that $f(5) = -30$, $f'(5) = 10$, $f''(5) = 16$ and $f'''(5) = 6$ and all higher derivatives of $f(x)$ at $x = 5$ are zero. 5+4+4 (CO2) (PO1)

9. SWE 19 FINAL

(PO1)

4. a) Thermistors are used to measure the temperature of bodies. These instruments are based on materials' change in resistance with temperature. To measure temperature, manufacturers provide you with a temperature vs. resistance calibration curve. If you measure resistance, you can find the temperature. Several observations with a thermistor are shown in Table 2. 11 (CO2) (PO1)

Page 1 of 2

Table 2: Resistance vs Temperature

R (ohm)	T (°C)
1101	25.113
911.3	30.131
636	40.12
451.1	50.128

Determine the temperature for 754.8 ohms using a 1st order Lagrange polynomial.

b) Using the previous data, determine the result for the same resistance (754.8 ohms) using a 2nd order Lagrange polynomial and find the absolute relative approximate error. 11+3 (CO3)

(PO2)

b) Taking upto 5 terms of the Taylor series, find the value of $\cos(\pi)$ given that we know $\cos\left(\frac{\pi}{2}\right) = 0$. 10 (CO2) (PO1)

10. SWE 19 QUIZ -2

2. The Taylor's series for the for e^x at point $x = 0$ is given by **6 (CO3) (PO2)**

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots$$

If we take more terms in the series, we get less truncation error and thus obtain a better estimate for e^1. Exactly how many terms do we need to take in order to get an error less than 0.001 in the approximation of e^1.

11. CSE 18 Quiz question

Department of Computer Science and Engineering (CSE)

Quiz no. 2 **Summer Semester 2020-2021**

DURATION: 30 MIN **FULL MARKS: 15**

Math 4641: Numerical Methods

Figures in the right margin indicate marks.

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1. Differentiate between the solution approaches of Direct, Newton's Divided Difference and Lagrangian Interpolation. **4 (CO1) (PO2)**

2. Suppose you are given n data points and you are trying to perform **Linear interpolation** at an unknown arbitrary point x. For Linear interpolation, you would need to choose two closest data points to x from the n given points so that these two points bracket the point x. Suppose these two points and their corresponding functional values are $(x_0, f(x_0))$ and $(x_1, f(x_1))$. **5 (CO1) (PO2)**

Now, if you wanted to perform the interpolation using Direct method, you would need to build a polynomial solving the unknown terms a_0 and a_1. Similarly, if you wanted to do the same task using Newton's Divided Difference method, then you would need to solve b_0 and b_1. Derive these parameters for each of the cases separately for linear interpolation. Also, show how you can get the parameters of Direct method from the parameters of Newton's Divided Difference method.

3. The upward velocity of a rocket is given as a function of time according to the following table: **6 (CO2) PO3)**

T (s)	V(t) (m/s)
0	0
10	227.04
15	362.78
20	517.35
22.5	602.97
30	901.67

Determine the velocity of the rocket at $t = 11$ seconds using 3rd order (cubic) Lagrangian polynomial interpolation. Also find the acceleration of the rocket at $t = 11$ as well as the distance travelled by the rocket from $t = 10$ to $t = 14$.

12. CSE 18 MID question

c) Explain the differences between **Interpolation** and **Extrapolation.** When would we need which? 6 (CO3) (PO2)

(PO2)

3 a) Suppose you are working at a start-up computer assembly company and you are tasked with determining the minimum number of computers the shop will have to sell to make a profit during the first year in business, which is also known as the **break-even point**. With such businesses, there is always some fixed capital cost, and some other costs which vary based on the number of computers assembled. At the break-even point, the total cost of operating the business and assembling the computers should become equal to the total sales minus the discount price. Based on this, suppose you found that the following equation should give you the break-even point: $\$35000 - 875n + 40n^{1.5} = 0$, where n is the number of computers sold. Find the minimum number of computers that need to be sold to obtain a profit by solving the above nonlinear equation using **Secant method**. Perform at least 3 iterations and show the number of significant digits that are at least correct in each iteration. 10 (CO2) (PO3)

b) Derive the Quadratic form of Newton-s Divided Difference polynomial and using its pattern, determine the Cubic form of Newton-s Divided Difference polynomial. 8 (CO1) (PO2)

(PO2)

c) The Maclaurin series is a special case of Taylor series. The Maclaurin series for $sin\ (x)$ is given as $sin\ sin\ (x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$. Use the **Remainder Theorem** to find the bounds of the truncation error in the representation of $sin\ (2)$ if only the first 3 terms of the Maclaurin series are used. 7 (CO2,CO3) (PO2)

13. CSE 18 Final question

c) Why do we normally use polynomials as interpolating functions? Can single polynomials always be used to accurately interpolate data? 2+5 (CO3) (PO2)

6. a) To maximize the catch of bass in a lake, it is suggested to throw the line to the depth of the thermocline. You can infer that there is a thermocline at a particular depth by noticing the sudden change in temperature. The following table provides us the depth vs temperature data for the lake: 6+7+3 (CO2, CO3) (PO3)

Table 1: Depth vs Temperature

$Depth, h(m)$	$Temperature, T$(°C)
0	19.1
-1	19.1
-2	19
-3	18.8
-4	18.7
-5	18.3
-6	18.2
-7	17.6
-8	11.7
-9	9.9
-10	9.1

It is noticed that there is a large difference in temperature from $h = -7$ to $h = -8$, so the thermocline must be somewhere between these two depths. The intermediate temperature data between these two depths is missing.

Determine the temperature at $h = -7.5$ using a **first order Lagrange polynomial** and then a **second order Lagrange polynomial** and find the absolute relative approximate error.

14. EXTRA QUESTION OF YEAR 17-18

Mid QUESTION OF YEAR 18

c) Given that $f(3) = 6$, $f'(3) = 8$, $f''(3)+1=12$ and that all other higher order derivatives of $f(x)$ are zero at x = 3, and assuming the function and all its derivatives exist and are continuous between x =3 and x = 7. Find out the value of f (7). 12

4. a) A robot arm with a rapid laser scanner is doing a quick quality check on holes drilled in a 15"x10" rectangular plate. The centers of the holes in the plate describe the path the arm needs to take, and the hole centers are located on a Cartesian coordinate system (with the origin at the bottom left corner of the plate) given by the specifications in Table 1. 12

Table 1: Dataset for Question 4.(a)

X (in)	Y (in)
2.00	7.2
4.25	7.1
5.25	6.0
7.81	5.0
9.20	3.5
10.60	5.0

If the laser is traversing from x = 2.00 to x = 4.25 to x = 5.25 in a quadratic path, what is the value of y at x = 4.00 using the direct method of interpolation and a second order polynomial?

MID QUESTION OF YEAR 17

be correct in the answer, then what you would need to do?

b) Answer the followings about Taylor series: 5+5

i. Why are the applications of Taylor's theorem important for numerical methods?

ii. Why the Taylor's Polynomial cannot be used for interpolation? 4

c) The Taylor series for e^x at point $x = 0$ is given by 2+8 O2) O3)

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots, \qquad -\infty < x < \infty$$

i. What is the truncation (true) error in the representation of $e^{2.5}$ if only four terms of the series are used?

ii. How many terms it would require to get an approximation of $e^{2.5}$ within a magnitude of true error of less than 10^{-6}?

42	4179
52	4186
82	4199
100	4217

Table 1: Specific heat C of water as a function of temperature T.

Determine the value of the specific heat at $T = 61°C$ using the ***Direct method*** of interpolation and a second order polynomial.

Solution:

For second order polynomial interpolation (also called quadratic interpolation), the equation for the specific heat would be –

$$C(T) = a_0 + a_1T + a_2T^2$$

Here, the order $n = 2$. Now, let's choose the nearest $n + 1 = 2 + 1 = 3$ points that bracket the point $T = 61°C$ and evaluate the polynomial. The 3 points are $T_0 = 42$, $T_1 = 52$, and $T_2 = 82$. The respective specific heat values are $C(T_0) = 4179$, $C(T_1) = 4186$, and $C(T_2) = 4199$. Using these values, we obtain the following equations –

$$C(42) = a_0 + a_1 \times 42 + a_2 \times (42)^2 = 4179$$
$$C(52) = a_0 + a_1 \times 52 + a_2 \times (52)^2 = 4186$$
$$C(82) = a_0 + a_1 \times 82 + a_2 \times (82)^2 = 4199$$

Writing these equations in matrix form, we have –

$$\begin{bmatrix} 1 & 42 & 1764 \\ 1 & 52 & 2704 \\ 1 & 82 & 6724 \end{bmatrix} \cdot \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 4179 \\ 4186 \\ 4199 \end{bmatrix} \Rightarrow \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 1 & 42 & 1764 \\ 1 & 52 & 2704 \\ 1 & 82 & 6724 \end{bmatrix}^{-1} \cdot \begin{bmatrix} 4179 \\ 4186 \\ 4199 \end{bmatrix}$$

$$\begin{bmatrix} a_0 \\ \end{bmatrix} \begin{bmatrix} \frac{533}{50} & -\frac{287}{25} & \frac{91}{50} \end{bmatrix} \begin{bmatrix} 4179 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 82 & 6724 \end{bmatrix} \quad \begin{bmatrix} a_2 \end{bmatrix} \quad \begin{bmatrix} 4199 \end{bmatrix} \qquad \begin{bmatrix} a_2 \end{bmatrix} \quad \begin{bmatrix} 1 & 82 & 6724 \end{bmatrix} \qquad \begin{bmatrix} 4199 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} \frac{533}{50} & -\frac{287}{25} & \frac{91}{50} \\ -\frac{67}{200} & \frac{31}{75} & -\frac{47}{600} \\ \frac{1}{400} & -\frac{1}{300} & \frac{1}{1200} \end{bmatrix} \cdot \begin{bmatrix} 4179 \\ 4186 \\ 4199 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 4135 \\ 1.3267 \\ -6.6667 \times 10^{-3} \end{bmatrix}$$

Hence,

$$C(T) = 4135 + 1.3267T - 6.6667 \times 10^{-3} T^2; \qquad 42 \leq T \leq 82$$

At the point $T = 61$, we get,

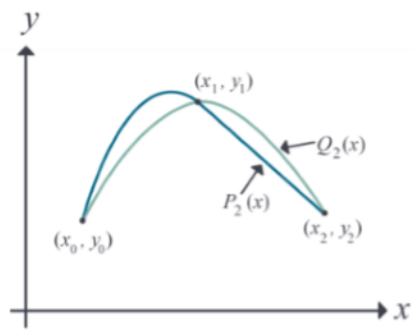
$$C(61) = 4135 + 1.3267 \times 61 - 6.6667 \times 10^{-3} \times (61)^2 = 4191.2 Jkg^{-1} {}^\circ C^{-1}$$

Rubric:

- 1 point for the correct polynomial representation.
- 1 point for choosing the correct nearest points.
- 1 point for the correct coefficient values.
- 1 point for the correct answer.

4. Prove that a polynomial of degree n or less that passes through $n+1$ data points is unique. 3 (CO1) (PO2)

Solution:



Proof: Let us use proof by contradiction. If the polynomial is not unique, then at least two polynomials of order n or less pass through the $n+1$ data points.

Assume two polynomials $P_n(x)$ and $Q_n(x)$ go through $n+1$ data points (x_0, y_0), (x_1, y_1), $\ldots$, (x_n, y_n). Then,

$$R_n(x) = P_n(x) - Q_n(x)$$

Since $P_n(x)$ and $Q_n(x)$ pass through all the $n+1$ data points, we can say,

$$P_n(x_i) = Q_n(x_i) \quad \forall i \in \{0, \ldots, n\}$$

Hence

$$R_n(x_i) = P_n(x_i) - Q_n(x_i) \quad \forall i \in \{0, \ldots, n\}$$
$$\Rightarrow R_n(x_i) = 0$$

The nth order polynomial $R_n(x)$ has $n+1$ zeros. A polynomial of order n can have $n+1$ zeros only if it is identical to a zero polynomial, that is,

$$R_n(x) \equiv 0$$
$$\Rightarrow P_n(x) \equiv Q_n(x)$$

So, $P_n(x)$ and $Q_n(x)$ must be the same unique polynomial, which contradicts our initial assumption. □ Q.E.D.

Rubric:

- 1 point for correctly defining $P_n(x)$, $Q_n(x)$, and $R_n(x)$.
- 1 point for showing $R_n(x)$ is a zero polynomial.
- 1 point for the final conclusion.

1. **CSE quiz-19**